

MODULE 2 · TOPIC 2

RTL Design Methodology

Basics of register transfer level (RTL) design · Overview of the RTL design process and methodology · Introduction to hardware description languages such as Verilog or VHDL

- Theory 1 · What RTL is — the two kinds of logic, the synchronous discipline, four levels of abstraction
- Theory 2 · The methodology — the flow, the synthesisable subset, the rules, and coding style
- Theory 3 · The patterns — datapath and controller, state machines, parameters and generate
- Theory 4 · HDLs — what one is, and Verilog against VHDL
- Practical · Labs A-N · 26 hours · 103 exercises · every number measured



Terminal Outcomes

Module 2 terminal outcomes

NOS NIE/ELE/N0102 · Verilog RTL coding for Synthesis · 25 h theory + 35 h practical

1

Understand the design cycle of VLSI

where RTL sits in the flow, and what happens either side of it

Topic 1 and 2

2

Understand Verilog syntax, LEVELS OF ABSTRACTION, and testbench simulation

the abstraction ladder is this subtopic's central idea

Topic 2, 4, 5

3

Design and develop IPs for VLSI using Verilog

which begins with knowing what is synthesisable and what is not

Topic 2, 4

4

Emulate, debug and characterise reusable IPs

reuse is a property you design in, not one you add later

Topic 2, 5, 6

Outcome 2 names this subtopic explicitly

"Level of abstraction in Verilog programming" is the phrase in the NOS. This session takes one circuit down all four levels, simulates them together, and proves they are the same.

That is the deliverable this subtopic exists to produce.

Outcome 2 names this subtopic in the NOS itself

"Level of abstraction in Verilog programming" is the phrase. This session takes one circuit down all four levels, simulates them together, and proves they are the same — which is the deliverable the outcome asks for.



Key Learning Outcomes

Key learning outcomes for this subtopic

THEORY

- Explain what register transfer level means, and the timing model it implies
- Describe the levels of abstraction in an HDL and choose between them
- Explain how timing constraints and the synthesisable subset shape the RTL you write
- Compare Verilog and VHDL

PRACTICAL

- Write RTL code for basic digital circuits
- Validate RTL designs through simulation using testbenches
- Apply a coding standard and check it mechanically
- Synthesise a design and read what the tool built

Every one of them is assessed by something you run

what RTL means -> make transfer, make ladder
levels of abstraction -> make ladder, make prove
the coding standard -> make lint, make lintcheck
synthesis -> make subset, make flow
Verilog or VHDL -> make langs

Every outcome is assessed by a command you run, not by something you recite.



Every Syllabus Phrase, and Where It Is Covered

Every syllabus phrase, and where it is covered

syllabus phrase	section	slides
Basics of register transfer level (RTL) design	Theory 1	6-25
Overview of RTL design process and methodology	Theory 2	26-41
... and the patterns that process produces	Theory 3	42-59
Introduction to hardware description languages (HDLs) such as Verilog or VHDL	Theory 4	60-72
Practical: RTL Design and Implementation Labs	Labs A-N	73-89

The syllabus gives this subtopic 4 notional hours

Four hours is enough to define the terms. It is not enough to make anyone believe them, and nothing in this subtopic is believed until it has been seen to happen.

So the theory is developed to the depth the ideas need, and every claim is attached to a lab target that produces it. Deliver it in four hours by lecturing from the summary slides; deliver it properly by running the lab alongside.

On the four notional hours

Four hours is enough to define the terms. It is not enough to make anyone believe them — and nothing in this subtopic is believed until it has been seen to happen. Lecture from the summary slides to fit four hours; run the lab alongside to teach it.



How This Session Runs

How this session runs

THEORY 1

What RTL is

combinational and sequential, the synchronous discipline, four levels of abstraction

THEORY 2

The methodology

the flow, the subset, the rules, and why each exists

THEORY 3

The patterns

datapath and controller, the finite state machine, parameters and generate

THEORY 4

HDLs

what an HDL is, and Verilog against VHDL

PRACTICAL

Labs A-N

26 hours, 103 exercises, all measured

One sentence to open with

RTL is not a language and it is not a tool. It is a way of thinking about hardware in which you say which registers exist and what transfers into them - and leave everything else to the tool.

Everything else in this session is a consequence of that sentence: what you may write, what you may not, why the rules exist, and what the tool does with it.

RTL is not a language and it is not a tool. It is a way of thinking about hardware.

THEORY 1

What RTL Is

The name is the definition, and almost everything else in this topic is a consequence of it.

- Registers, and what transfers into them on each clock edge
- Why `<=` and `=` are not interchangeable
- Four levels of abstraction, one circuit
- And what a synthesiser makes of each level



Register Transfer Level: the Name Is the Definition

Register Transfer Level: the name is the definition

An RTL description says two things, and nothing else.

1. WHICH REGISTERS EXIST

2. WHAT TRANSFERS INTO EACH, ON EACH CLOCK EDGE



```
always @(posedge clk) begin
    y <= x + 1;    // x, through an adder, into y
end
```

Everything else is the tool's problem

How many gates, which gates, how they are wired, how fast they are - none of that is in the description. You state the transfers; synthesis works out the circuit.

Two statements, and nothing else

1. Which registers exist. 2. What transfers into each one, on each clock edge.

How many gates, which gates, how they are wired, how fast they are — none of that is in the description. You state the transfers; synthesis works out the circuit.



Watch One Value Transfer

Watch one value transfer, one register per edge

A single 5 is applied on cycle 0 and never again.

cycle	din	x	y	z	acc
0	5	5	1	0	0
1	0	0	6	2	0
2	0	0	1	12	2
3	0	0	1	2	14
4	0	0	1	2	16

5 -> x 6 -> y 12 -> z -> acc
one edge one edge one edge

Nothing moved between edges

The logic settles during the cycle and the registers all update together at the edge. That is the entire timing model of RTL, and it is why RTL is easy to reason about.

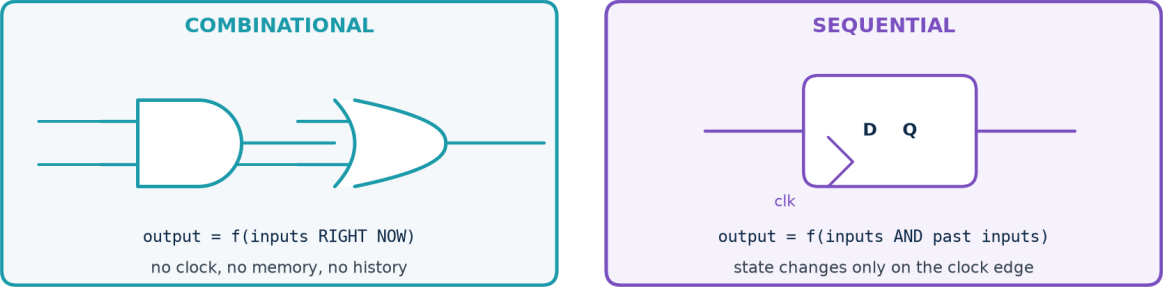
A single 5 applied on cycle 0. It reaches x, then y as 6, then z as 12, then acc — one register per clock edge, and nothing moves in between.



Combinational and Sequential

The only two kinds of digital logic there are

Every RTL block you will ever write is some arrangement of these two.



	combinational	sequential
what it is	logic gates	flip-flops (and the logic feeding them)
Verilog	always @* ... =	always @(posedge clk) ... <=
what it remembers	nothing	one value per flip-flop
what can go wrong	an inferred latch, a glitch	a setup or hold violation
where it is checked	simulation, lint	simulation, lint, STA

The whole of RTL design is deciding what goes in the registers and what happens between them

That is not a slogan - it is literally what the letters R, T and L stand for.

There is no third kind. Every block in this course is an arrangement of these two.



The Same Distinction, In Code

COMBINATIONAL

always @(*) or assign

Blocking assignment. Every output written on every path, or the tool builds a latch you did not ask for.

No clock appears anywhere in the block.

Checked by: simulation, lint.

SEQUENTIAL

always @(posedge clk)

Non-blocking assignment. Reset appears here and only here.

One value remembered per flip-flop, updated on the edge and at no other time.

Checked by: simulation, lint, and static timing analysis.

Why the two are kept in separate always blocks

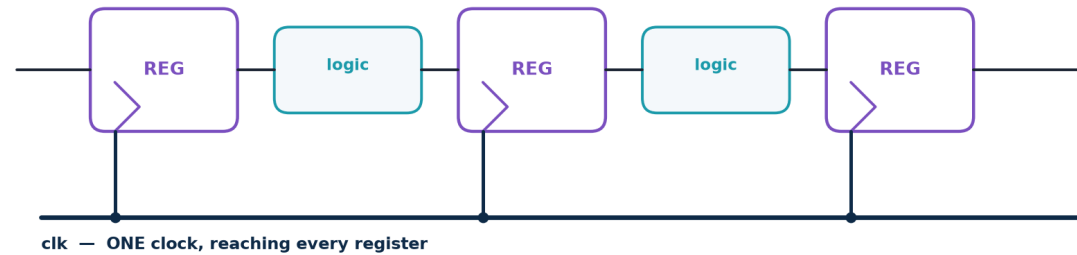
A block that mixes them has to be read twice — once for what it computes and once for when. Splitting them means each block has one job, and the reviewer, the linter and the synthesiser all agree about which job it is.



One Clock, One Edge, Everything

The synchronous discipline: one clock, one edge, everything

Almost every rule in this topic is a consequence of this one decision.



the rule	what it means	why
one clock edge	every register samples at the same instant	so 'now' means the same thing everywhere
one reset policy	sync or async, active high or low - pick one	so nobody has to look it up per module
no logic on the clock	no gated clocks, no ripple clocks	so STA has one thing to analyse
no latches	assign every output on every path	so timing is edge-to-edge, not level-dependent
registered outputs	a block's outputs come out of flops	so a slow path never crosses two blocks

You are allowed to break these rules. You are not allowed to break them by accident.

Clock gating and multiple clock domains are real techniques - deliberate, reviewed and constrained.
That is a different thing from a design that drifted out of the discipline without anyone deciding.

Almost every rule in this topic is a consequence of this one decision.



What the Discipline Buys You

ANALYSABLE

With one clock and one edge, timing analysis is a finite question: for every path from a flip-flop to a flip-flop, does the data arrive in time? Add a gated clock and the question multiplies.

COMPOSABLE

Two blocks written to the same discipline can be wired together without a conversation. Two blocks written to different disciplines need one, every time.

REVIEWABLE

A reviewer who knows the discipline can read your block for what it computes, because the question of WHEN has already been answered the same way it always is.

TESTABLE

Scan insertion, the technique that makes a chip testable after manufacture, assumes edge-triggered flip-flops on one clock. Latches and gated clocks each need special handling.

You are allowed to break these rules. You are not allowed to break them by accident.



Why Anyone Designs At This Level

	Behavioural / algorithmic	RTL	Gate netlist
you write	the algorithm	registers and transfers	every gate
timing	none at all	one clock period per stage	exact, per gate
synthesisable	rarely	yes — this is the target	yes, but why would you
a 10k-gate design	unbuildable	a few hundred lines	tens of thousands of lines
who writes it	architects, in C or SystemC	you	the synthesiser

RTL is the level where the trade lands correctly

High enough that a human can write and read a real design; low enough that a tool can build it without guessing at your intent.

Every industrial digital design in the last thirty years was written here. That is not fashion — it is where the abstraction pays for itself.

Above RTL you cannot say when things happen. Below it you cannot say anything else.



The Two Assignment Operators, and Why It Matters

Why clocked blocks use <= and combinational blocks use =

<= non-blocking

in CLOCKED blocks
every right-hand side is read first,
using the OLD values,
then every left-hand side updates
at the same instant

`a <= b; b <= a;` swaps them.

= blocking

in COMBINATIONAL blocks
each statement completes before
the next one starts,
exactly the way software
statements run

`a = b; b = a;` does not.

Get it backwards and the code still compiles

Blocking assignments in a clocked block make two blocks see each other's half-updated values, and which one wins depends on the order the simulator happens to run them in.

Non-blocking in a combinational block makes it behave like a register in simulation while synthesis builds plain logic - so the simulation and the silicon disagree.

Neither one is an error. Both are caught by rules L001 and L002 of the linter in this lab.

Neither one is an error. Both are caught by the linter you build in this topic.



The Swap, Worked Through

```
// NON-BLOCKING - inside always @(posedge clk)
// Step 1: read every right-hand side, using the OLD values.
// Step 2: update every left-hand side, all at the same instant.

    a <= b;      // reads old b
    b <= a;      // reads old a

    a and b are SWAPPED. This is what a hardware register does.

// BLOCKING - inside always @*
// Each statement finishes before the next one starts.

    a = b;      // a is now b
    b = a;      // a is already b, so this does nothing

    BOTH end up holding b. This is what software does.
```

Why the rule is absolute rather than stylistic

Two clocked blocks using blocking assignments can see each other's half-updated values, and which one wins depends on the order the simulator happens to evaluate them in — an order the language standard deliberately does not fix.

Non-blocking assignment exists precisely to make that race impossible.



And What Each One Actually Builds

Blocking in a clocked block: what it actually builds

The same three lines. One character different on each. make pitfalls

shift_nb — correct

```
always @(posedge clk) begin
    q[0] <= din;
    q[1] <= q[0];
    q[2] <= q[1];
end
```

3 flip-flops

a 3-stage delay line

shift_bl — not a shift register

```
always @(posedge clk) begin
    q[0] = din;
    q[1] = q[0];
    q[2] = q[1];
end
```

1 flip-flop

din reaches q[2] in ONE cycle

cycle	din	q_nb	q_bl	expected q[2]
2	0	100	000	1
4	1	001	111	0
6	0	110	000	1

non-blocking: 0 wrong cycles · blocking: 6 wrong cycles out of 10 checked

It compiled. It simulated. It synthesised. No tool issued a single warning, because nothing illegal was written - the code simply describes a different circuit from the one the author had in mind.

This is what a methodology rule is FOR. Rule L001 in the lab linter catches it in about a millisecond.

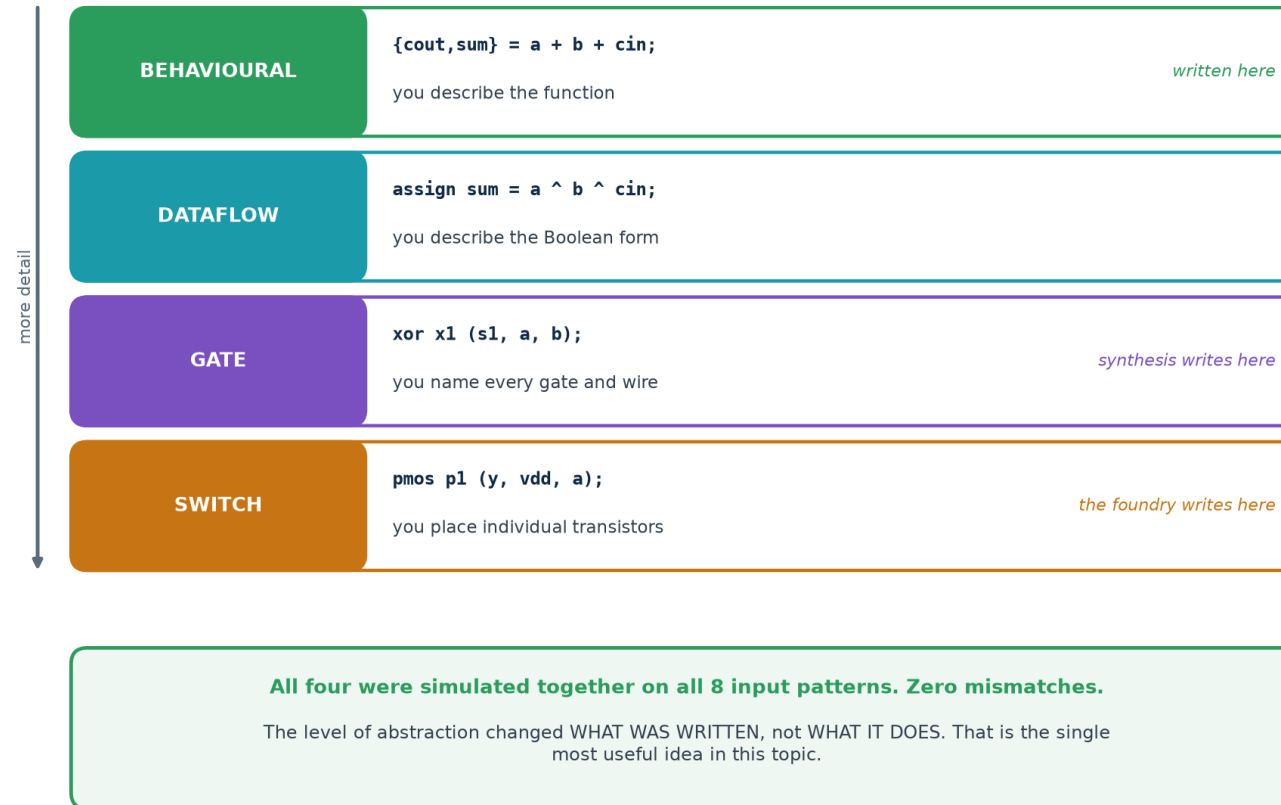
Three flip-flops against one. Nothing illegal was written, so nothing warned.



Four Levels, One Circuit

Four levels of abstraction, one circuit

Higher up you say WHAT. Lower down you say HOW. The circuit is the same.



All four were simulated together on all 8 input patterns. Zero mismatches.



The Same Adder, Written Four Ways

```
// BEHAVIOURAL - you describe the function
always @* {cout, sum} = a + b + cin;

// DATAFLOW - you describe the Boolean form
assign sum = a ^ b ^ cin;
assign cout = (a & b) | (b & cin) | (a & cin);

// GATE - you name every gate and every wire
xor x1 (s1, a, b);      xor x2 (sum, s1, cin);
and a1 (ab, a, b);      and a2 (bc, b, cin);  and a3 (ac, a, cin);
or  o1 (cout, ab, bc, ac);

// SWITCH - you place individual transistors
pmos p1 (y, vdd, a);    nmos n1 (y, gnd, a);    // one CMOS inverter
```



And What a Synthesiser Makes of Each

And what a synthesiser makes of each level

written at	synthesis	cells	what came out
behavioural	OK	5	NAND x3, XOR x2
dataflow	OK	6	AND x2, OR x2, XNOR, ORNOT
gate	OK	6	identical netlist to dataflow
switch	REFUSED	-	transistors are not synthesisable

Two results worth stopping on

The BEHAVIOURAL description produced the SMALLEST circuit - five cells against six.
And dataflow and gate produced the IDENTICAL netlist: once you write the Boolean expression you have already chosen the structure, and naming the gates adds nothing but typing.

The rule this gives you

Write at the highest level that expresses your intent. Every level you descend takes a decision away from the tool and gives it to you - whether or not you wanted it.

The tool is better than you are at choosing gates. It is not better than you are at choosing architecture.

The behavioural description produced the SMALLEST circuit — and dataflow and gate produced the identical netlist.



The Rule This Gives You

Write at the highest level that expresses your intent

Every level you descend takes a decision away from the tool and gives it to you — whether or not you wanted it.

What the tool is better at than you

Choosing a gate mix for a Boolean function.

Balancing a logic tree.

Meeting a timing constraint by restructuring.

Doing all of that again, consistently, at 3 a.m.

What you are better at than the tool

How many clock cycles the job should take.

Where the registers go.

What is shared and what is duplicated.

What the interface should look like.

Give the tool the first list. Keep the second one for yourself.



Simulation Shows. Proof Settles.

Simulation shows. Proof settles.

EXHAUSTIVE SIMULATION

8 patterns, 3 inputs.
Every possible case, so this
really is complete.

*but 2^{30} patterns is not,
and real designs are bigger*

EQUIVALENCE CHECKING

Build a miter: both designs,
same inputs, outputs compared,
assert they always agree.

*A solver then tries to break it
and cannot. No enumeration at all.*

proved in the lab

behav vs dataflow EQUIVALENT (94 SAT variables)

behav vs gate EQUIVALENT (94 SAT variables)

behav vs broken NOT EQUIVALENT

That last line is the important one

fa_broken has one carry term missing, so it is wrong for one input pattern in eight. A checker that cannot fail is not evidence of anything - you have to watch it catch a real bug.

A checker that cannot fail is not evidence of anything — which is why the lab includes a deliberately broken adder.

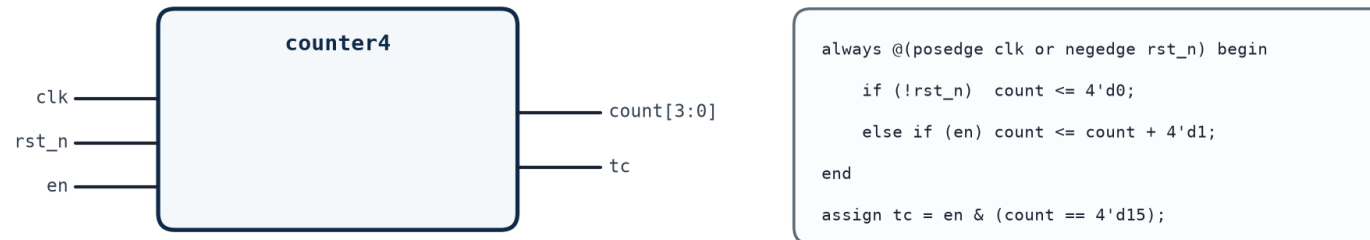


1.8 · THE RUNNING EXAMPLE

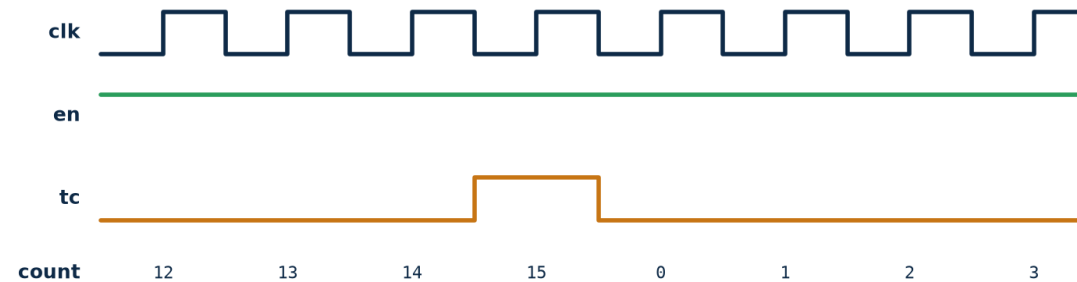
One Design, Carried All the Way Through

The running example: a 4-bit counter

One design, carried through the coding, the testbench, the flow and the reuse lab.



tc is high for exactly one cycle, while count == 15



The same design comes back three times

Lab 3 runs it through the whole front end · Lab 4 turns its controller into an FSM · Lab 7 parameterises it and instantiates it twice

Four bits, an asynchronous reset, an enable and a terminal count. Small enough to read in one glance, big enough to be a real design.



Read It Line By Line

```
module counter4 (  
    input        clk,  
    input        rst_n,    // asynchronous, active LOW  
    input        en,  
    output reg [3:0] count,  
    output        tc        // one cycle high at 15  
);  
    always @(posedge clk or negedge rst_n) begin  
        if (!rst_n)    count <= 4'd0;    // reset wins  
        else if (en)    count <= count + 4'd1;  
    end                // no else: HOLD  
  
    assign tc = en & (count == 4'd15);  
endmodule
```

Four decisions are visible in fourteen lines

The reset is ASYNCHRONOUS, so it is in the sensitivity list; it is ACTIVE LOW, so the port is named `rst_n` and the test is `!rst_n`. Reset is tested FIRST, so it wins over the enable. And the missing else is deliberate — in a CLOCKED block, no assignment means hold, which is what a flip-flop does anyway.



The Numbers, Worked Out

A worked numerical example

The counter and the accumulator, with the arithmetic written out.

counter4 at 100 MHz

clock period	$1 / 100 \text{ MHz} = 10 \text{ ns}$
states before it wraps	$2^4 = 16$
time for one full cycle	$16 \times 10 \text{ ns} = 160 \text{ ns}$
tc pulse rate	$1 / 160 \text{ ns} = 6.25 \text{ MHz}$
tc pulse width	$10 \text{ ns} - \text{exactly one clock}$

accum_top, N = 6 samples

cycle 0	start seen, sum cleared, cnt <- 6
cycles 1..6	one sample added per clock
cycle 7	cnt reached 0, machine moves on
cycle 8	done pulses, sum is valid
total latency	$N + 3 = 9 \text{ clocks}$

Why the latency formula is worth deriving rather than measuring

One cycle to accept the request, N to accumulate, one to notice the counter hit zero, one to flag it. That is $N + 3$, and it holds for every N - so a testbench that only ever tries $N = 6$ has told you almost nothing.

The measured run agrees: sum = 157 for samples 10+20+30+40+50+7, done at cycle 8.

Throughput, not latency, is what usually matters

This design accepts one sample per clock while it is running: 100 M samples/s at 100 MHz.
The three extra cycles are paid once per batch, not once per sample.

Derive the formula, then measure it. A testbench that only ever tries one value of N has told you almost nothing.



Nine Questions

#	Question	The answer in one line
1	What two things does an RTL description state?	which registers exist, and what transfers into them
2	What does <code><=</code> mean inside a clocked block?	read all right-hand sides first, then update together
3	Name the four levels of abstraction.	behavioural, dataflow, gate, switch
4	Which level produced the smallest netlist, and why?	behavioural — it left the tool free to choose
5	Why did dataflow and gate give identical netlists?	writing the Boolean form already fixes the structure
6	Why prove equivalence instead of simulating?	exhaustive simulation stops being possible at ~30 inputs
7	What are the only two kinds of digital logic?	combinational and sequential — there is no third
8	Name three things the synchronous discipline buys you.	analysable timing, composable blocks, testable silicon
9	In counter4, why is there no else on the enable?	in a clocked block, no assignment means hold

Theory 2 asks what you are allowed to write, and why the rules exist.

THEORY 2

The RTL Design Process and Methodology

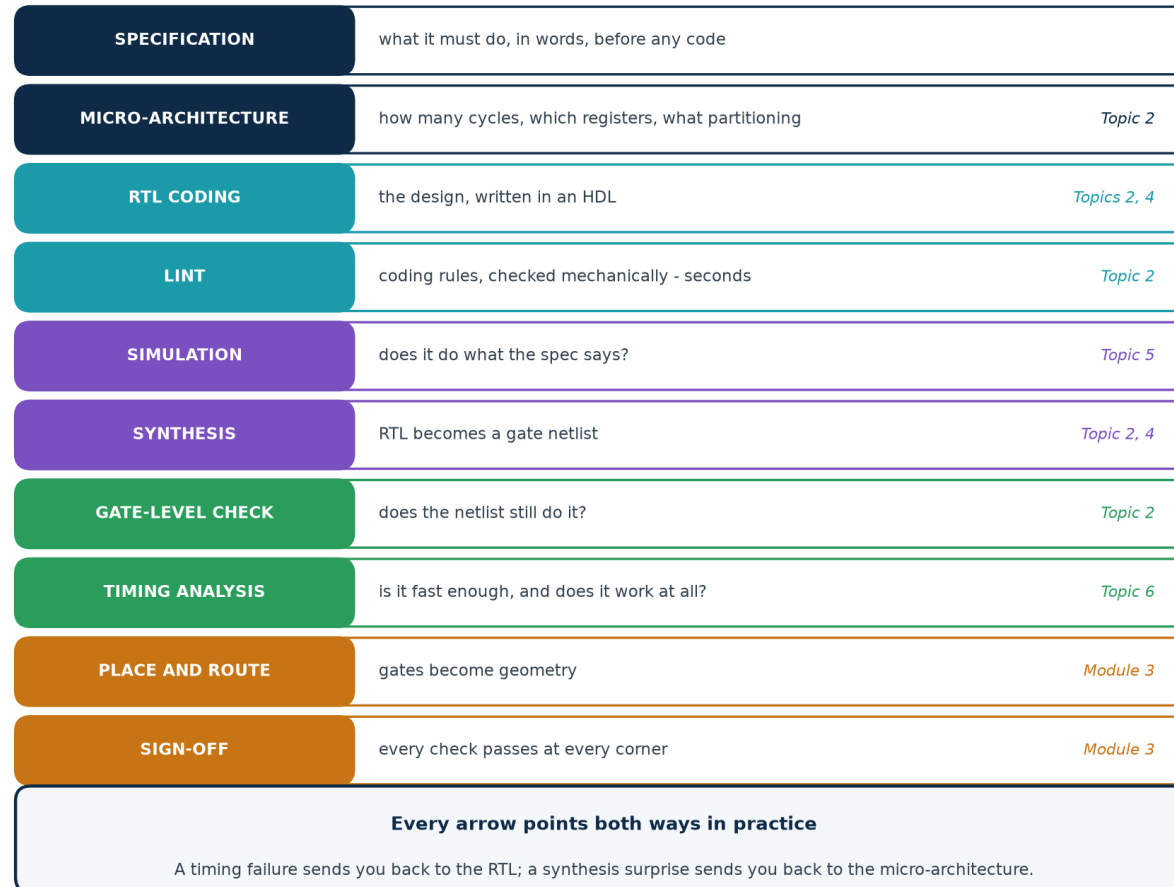
Methodology is the hardest thing to teach, because the honest version is a list of habits and the dishonest version is a list of slogans.

- The design flow, and where this topic sits in it
- The synthesisable subset — measured, not asserted
- The bug that makes simulation and silicon disagree
- Seven coding rules, and a tool that checks them
- Micro-architecture: the decisions RTL cannot make for you



The RTL Design Flow

The RTL design flow, and where this topic sits



Every arrow points both ways in practice: a timing failure sends you back to the RTL, a synthesis surprise back to the micro-architecture.



2.1 · THE FLOW

Executed, Not Described

A methodology is a set of gates, not a set of suggestions

make flow - seven stages on one design, each producing evidence.

	stage	what runs	evidence produced
1	SPEC	written before the RTL	4 sentences
2	LINT	tools/rtl_lint.py	0 Issues
3	RTL SIMULATION	iverilog, 18 cycles	wraps at 15, tc correct
4	SYNTHESIS	yosys	12 cells
5	GATE SIMULATION	the netlist, same stimulus	18 cycles
6	COMPARE	RTL against gate transcript	identical
7	PROVE	formal equivalence	proven by induction

Stage 7 is the one worth understanding

Stage 6 says the two agree on the 18 cycles that were tested. Stage 7 proves they agree on EVERY input sequence, by induction, without enumerating any of them.

The script stops at the first stage that fails. That is what makes it a methodology.

make flow stops at the first stage that fails. That is what makes it a methodology rather than a diagram.



Which Verilog Actually Synthesises

Which Verilog actually synthesises - measured

"Verilog is not a programming language" stays a slogan until you watch a tool refuse.

construct	synth	cells	what the tool said
always @* with full case	OK	10	clean
clocked block with <=	OK	8	8 flip-flops
if with no else	OK	1	inferred a LATCH
incomplete sensitivity list	OK	1	built anyway - see next slide
#5 delay in RTL	OK	1	the delay silently vanished
initial block	OK	10	accepted - FPGA only, not ASIC
for loop, constant bounds	OK	7	unrolled into 7 gates
while loop, data-dependent	REFUSED	-	"only allowed in constant functions"
real (floating point)	REFUSED	-	"syntax error, unexpected TOK_REAL"
a / b (variable divisor)	OK	371	a full combinational divider
a / 4 (constant divisor)	OK	0	no logic at all - just wires

371 cells against 0, for the same operator

The difference is entirely what you divided BY. A constant power of two is a rename of wires;
a variable divisor is one of the largest things you can ask for by accident.

Eleven constructs, run through a real synthesiser. The table is measured, not remembered.



Three Rows Worth Discussing

371 cells against 0

`a / b` where `b` is a signal builds a full combinational divider — 371 cells.

`a / 4` builds nothing at all. It is a rename of wires.

Same operator. The difference is entirely what you divided BY.

#5 vanished silently

A delay is a simulation instruction. Silicon has no way to wait five time units.

Synthesis ignored it and built the gate. Your simulation and your chip now behave differently.

initial was accepted

Yosys targets FPGAs, where the bitstream really does initialise the flops.

An ASIC flow would not. This is how code that works on an FPGA fails on an ASIC.

The habit this table is for

Before you write a construct you are unsure about, synthesise a ten-line module containing only that construct and read what came out. It takes two minutes and it is the only way to actually know.



2.3 · THE LATCH

The Most Common RTL Bug There Is

The inferred latch: the most common RTL bug there is

```
always @* begin
    if (en) y = d;
end
```

there is no else, so you never said what y is when en = 0

so the tool must build something that REMEMBERS

\$_DLATCH_P_

a level-sensitive latch you did not ask for

Why it is worse than an error

It is not an error at all. The tool builds it, mentions it in a log nobody reads, and hands you a design with a memory element in the middle of what you thought was combinational logic - which then has its own timing requirements, and breaks your static timing analysis.

It is not an error. The tool builds it, mentions it in a log nobody reads, and hands you the design.



Why a Latch In the Middle of Your Logic Hurts

	A flip-flop	A latch
captures	on the clock EDGE	while the enable is HIGH
is transparent	never	for the whole enable window
timing analysis	one clean check per edge	time-borrowing; much harder to analyse
in a scan chain	standard	usually needs special handling
you asked for it	yes, by writing posedge	no — you forgot an else

So the rule is mechanical

In a combinational block, assign every output on every path. An if needs an else; a case needs a default. If you would rather not write them out, assign a default value at the top of the block and let later assignments override it.

Rules L005 and L006 of the linter catch this, and make lintcheck confirms the linter agrees with Yosys on every file.



When Simulation and Silicon Disagree

When simulation and silicon disagree

```
always @(a) begin y = a & b; end - b is read, but not in the list.
```

stimulus	a	b	RTL y	NETLIST y	
start	0	0	0	0	
change a	1	0	0	0	
change b	1	1	0	1	DISAGREE
change b	1	0	0	0	
change a	0	0	0	0	
change b	0	1	0	0	

Your testbench was verifying a circuit that will never be built

The RTL only re-evaluates when a changes, so y goes stale whenever b moves alone.
Synthesis ignored the sensitivity list entirely and built the AND gate you meant.

One disagreement in six - and nothing anywhere reported an error.

Which is why the rule is: never maintain a sensitivity list by hand

always @* builds it for you, and keeps it correct every time you edit the block.
SystemVerilog's always_comb does the same and also makes the tool check it.

One disagreement in six — and nothing anywhere reported an error.



Why This Class of Bug Is Special

Most bugs are found by testing. This one cannot be, because **the thing you are testing is not the thing that will be built**. Every test you write passes, and the chip still fails.

Cause	What simulation does	What synthesis does
incomplete sensitivity list	the block sleeps; the output goes stale	ignores the list; builds the logic
# delays in RTL	waits the specified time	ignores them entirely
initial block (ASIC)	sets the value at time zero	ignores it; the flop powers up unknown
blocking in a clocked block	result depends on evaluation order	builds one specific circuit

All four have the same cure

Use always @* and never a hand-written sensitivity list. No delays in RTL. A real reset instead of an initial block. Non-blocking in clocked blocks. Then lint for all four, so that nobody has to remember.



Seven Rules, Checked By a Tool

Seven rules, checked by a tool instead of by memory

rule	what it catches	why it matters
L001	blocking (=) in a clocked block	two blocks see half-updated values
L002	non-blocking (<=) in a combinational block	simulates like a register, synthesises as logic
L003	= and <= mixed in one block	no reader can tell the intent
L004	explicit sensitivity list	you have to maintain it, for ever
L005	if with no else in a combinational block	infers a LATCH
L006	case with no default	infers a LATCH
L007	signal driven from two always blocks	two drivers - last one to run wins

And the two latch rules are not opinions

L005 and L006 claim a synthesiser will build a latch. That is a claim about what a tool does, so the tool settles it: make lintcheck runs both and compares.

10 files · linter and Yosys agree on every one · 0 disagreements

A linter that cries wolf gets switched off; one that stays quiet is worse than none at all.

A linter that cries wolf gets switched off. One that stays quiet is worse than none at all.



The Coding Standard, In One Place

The RTL coding rules, in one place

Combinational logic

- always @* - never a hand-written sensitivity list
- blocking assignments (=) only
- assign every output on every path - else, or default

Sequential logic

- always @(posedge clk) - one clock, one edge
- non-blocking assignments (<=) only
- one signal, one always block, for ever

Never in synthesisable RTL

- # delays, initial blocks, real, unbounded loops
- both clock edges in the same path
- a clock built from combinational logic - use an enable

None of these are style preferences

Every one of them exists because breaking it produces a design that simulates differently from the way it is built - and that class of bug survives every test you write.

None of these are style preferences. Every one exists because breaking it produces a design that simulates differently from the way it is built.



The Decisions RTL Cannot Make For You

Micro-architecture: the decisions RTL cannot make for you

Synthesis chooses gates. It does not choose any of these.

the decision	the options	what it costs you
How many clock cycles?	one big cycle, or a pipeline	Fmax, latency, area
Where do the registers go?	stage boundaries	Fmax, and directly
What is shared?	one multiplier reused, or four in parallel	area against throughput
How is it partitioned?	module boundaries, hierarchy	readability, reuse, synthesis runtime
What is the interface?	handshake, valid/ready, fixed latency	how easily it plugs into anything else
Where does reset reach?	everything, or only what needs it	area, and routing congestion

This is where the engineering is

A synthesiser is better than you are at choosing gates. It is not better than you are at any of the rows above, and it will not warn you that you chose badly - it will faithfully build what you asked for.

Make these decisions on paper, before the first line of RTL.

Make these on paper, before the first line of RTL. A synthesiser will faithfully build whatever you chose.



Writing RTL That Someone Else Can Use

Writing RTL that someone else can use

Module 2's terminal outcomes ask for reusable IP. Reuse is a property you design in.

Parameterise, do not duplicate

WIDTH as a parameter, not four copies of the file

One clock, stated in the name

clk, and a comment saying which domain it belongs to

Reset policy, written down

synchronous or asynchronous, active high or low - pick one and keep it

No magic numbers

localparam with a name beats 8'd47 buried in a case

An interface, not a pile of wires

valid/ready is understood everywhere; your own scheme is not

A testbench that ships with it

IP without a testbench is a liability, not an asset

The test for reusable IP

Could a colleague drop it into a different design, next year, without asking you anything?
If the answer needs a conversation, it is not reusable yet.

Module 2's fourth terminal outcome asks for this by name

"Emulate, debug and characterise reusable IPs." Reuse is not something you add to a design afterwards — it is a set of decisions you take while writing it, and every one of them is listed above.



One Function, Three Coding Styles

One function, three coding styles - and three netlists

A 4:1 multiplexer. All 64 patterns simulated; every pair proved equivalent by SAT.

mux4_assign

```
assign y =  
  sel[1] ? (sel[0]?d[3]:d[2])  
        : (sel[0]?d[1]:d[0]);
```

3 cells

mux4_if

```
if      (sel==2'b00) y = d[0];  
else if (sel==2'b01) y = d[1];  
else ... else      y = d[3];
```

6 cells

mux4_case

```
case (sel)  
  2'b00: y = d[0]; ...  
  default: y = 1'b0;  
endcase
```

10 cells

EQUIVALENT is not the same as IDENTICAL

The SAT proof says all three compute the same Boolean function. It says nothing at all about area or timing, and on Yosys 0.33 the three netlists are 3, 6 and 10 cells.

The conditional expression uses sel[1] and sel[0] STRAIGHT as mux selects. The other two build equality comparators and ask the tool to re-derive that those comparisons are the select bits. It does not, quite.

So which do you write?

Whichever reads best - and then MEASURE, on your own tool, if the block is on a critical path. Folklore about what optimisers do is the least reliable knowledge in this field.

Proved equivalent by SAT — and still three different netlists.



What To Take From That

EQUIVALENT is not IDENTICAL

A formal equivalence proof answers exactly one question: do these two descriptions compute the same function?

It says nothing about area, nothing about timing and nothing about power. Those are separate questions with separate tools.

MEASURE YOUR OWN TOOL

The claim 'the optimiser flattens the difference' is the most repeated statement in RTL teaching, and on this toolchain it is false: 3, 6 and 10 cells.

A different synthesiser may well close the gap. Find out rather than assume.

SO WHICH DO YOU WRITE?

Whichever reads best to the next person, which is usually the case statement for anything with more than three branches. Then measure, and only reach for the terse form if the block is actually on a critical path.

Why the three differ at all

The conditional expression hands sel[1] and sel[0] straight to the multiplexer selects. The case and if/else versions ask the tool to build equality comparators against 2'b00, 2'b01 and so on, and then to work out for itself that those comparisons ARE the select bits. It gets part of the way.



Nine Questions

#	Question	The answer in one line
1	Name the first four stages of the flow.	spec, micro-architecture, RTL coding, lint
2	Why lint before simulating?	it costs seconds and catches what tests cannot
3	What builds a latch?	a combinational block that does not assign on every path
4	a / b against a / 4 — what did they cost?	371 cells against 0
5	Why is an incomplete sensitivity list so dangerous?	simulation and synthesis then build different circuits
6	What does # do in synthesisable RTL?	nothing at all — it is silently ignored
7	Name three decisions synthesis will not make for you.	cycle count, register placement, sharing, interface
8	Does formal equivalence say anything about area?	no — equivalent is not the same as identical
9	The three mux styles: how many cells each?	3, 6 and 10 — style did reach the netlist here

Theory 3 is about the patterns every real block is built from.

THEORY 3

The Patterns Every Block Is Built From

Three structures account for almost all synthesisable RTL. Once you can recognise them, most designs stop looking unfamiliar.

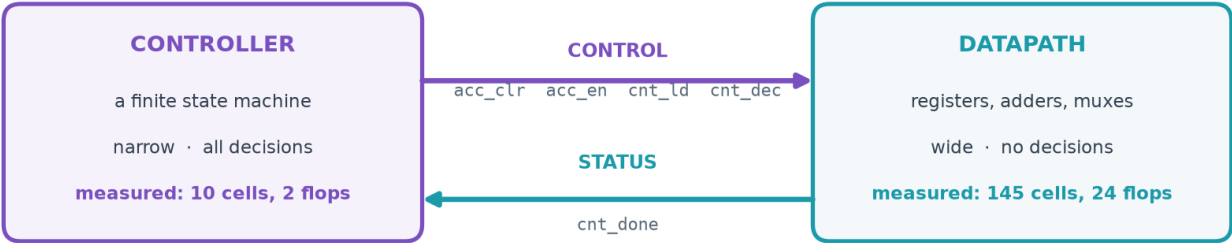
- Datapath and controller — the split that organises everything else
- The finite state machine, and the three-block way to write one
- Moore against Mealy, in structure and in measured timing
- State encoding, and what it actually costs
- Parameters, hierarchy and generate: from a module to an IP



Datapath and Controller

Datapath and controller: the shape of almost every block

A UART, a cache, a GPU. Different sizes, same two halves.



Two thin bundles of wires. That is the entire interface between the two halves.

when...	...this is why it is cheap
you re-time the datapath	the state machine does not change
you re-specify the sequence	no adder is touched
you widen the data	the controller is width-independent
you review the design	one half is 10 cells and can be read

`make dpctrl` – the accumulator, its FSM, and every control signal printed per cycle

The controller is 6% of the cells and 100% of the behaviour. That ratio is the reason for the split.

A UART, a cache, a GPU. Different sizes, the same two halves.



The Worked Example: Accumulate N Samples

```
// the CONTROLLER decides. Narrow. No data passes through it.
always @(*) begin
    acc_clr = 0; acc_en = 0; cnt_ld = 0; cnt_dec = 0; done = 0;
    case (state)
        S_IDLE : if (start)    begin acc_clr=1; cnt_ld =1; end
        S_RUN  : if (!cnt_done) begin acc_en =1; cnt_dec=1; end
        S_DONE :               done  =1;
    endcase
end

// the DATAPATH holds and transforms. Wide. It decides nothing.
always @(posedge clk or negedge rst_n)
    if      (!rst_n) sum <= 0;
    else if (acc_clr) sum <= 0;
    else if (acc_en)  sum <= sum + data;
```

Read the interface between them

Four control wires down, one status wire up. That is the entire contract. The controller never touches sum; the datapath never looks at state.



Measured, On the Accumulator

module	cells	flip-flops	what it contains
accum_ctrl	10	2	three states, five control outputs
accum_datapath	145	24	a 16-bit accumulator, an 8-bit down-counter
accum_top	156	26	both, plus the wires between them

Six per cent of the cells, one hundred per cent of the behaviour

That ratio is the whole argument for the split. You will re-time, widen and pipeline the expensive half many times over the life of a design — and each time, the half that decides what happens stays exactly as it was.

It also means the part that is hardest to get right is the part small enough to read in one sitting.

make dpctrl prints every control signal on every cycle, next to the accumulated sum.



The Three-Block Coding Pattern

How to write a state machine: the three-block pattern

Every FSM in this lab, and in most RTL groups, is written exactly like this.

BLOCK 1

state register

```
always @(posedge clk or negedge rst_n)
    if (!rst_n) state <= S_IDLE;
    else      state <= next_state;
```

sequential · <= · reset lives HERE and nowhere else

BLOCK 2

next-state logic

```
always @(*) begin
    next_state = state;    // the default
    case (state) ... endcase
end
```

combinational · = · the default assignment kills the latch

BLOCK 3

output logic

```
always @(*) begin
    out = 1'b0;           // defaults again
    case (state) ... endcase
end
```

combinational · reads STATE only if the machine is Moore

Why three blocks and not one

You can write a working FSM in a single clocked block. Three blocks are still better, because each one has exactly one job: block 1 is the only sequential logic, so it is the only place a reset or a clock appears; blocks 2 and 3 are pure functions of the state, so they can be read, reviewed and changed without thinking about time at all.

`make fsm` — three machines written this way, simulated and synthesised

Every state machine in this lab is written exactly like this. So is most industrial RTL.



Why Not One Block?

You can write a working state machine in a single clocked block, with the case statement inside it. Nothing about it is illegal.

	one block	three blocks
works?	yes	yes
where is the reset?	mixed in with the logic	block 1, and nowhere else
where is the clock?	wrapped around everything	block 1, and nowhere else
outputs are	registered, always	your choice — Moore or Mealy
reviewing a change	you re-read the timing every time	blocks 2 and 3 have no timing to re-read
adding a state	one place, easy to miss a branch	two places, both of them a case statement

One block is not wrong. Three blocks are easier to be right in, which over a project is the same thing.



The Default Assignment, and Why It Matters

```
always @(*) begin
    next_state = state;          // <-- THE DEFAULT ASSIGNMENT
    case (state)
        S_MAIN_GREEN : if (car && timeout) next_state = S_MAIN_YELLOW;
        S_MAIN_YELLOW : if (timeout)      next_state = S_SIDE_GREEN;
        ...
    endcase
end

// Not one of those if statements has an else. Not one of them needs
// one: next_state was already written, unconditionally, on line 2.
// No path through this block leaves it unassigned, so no latch.
```

This is the idiom the lab linter had to learn

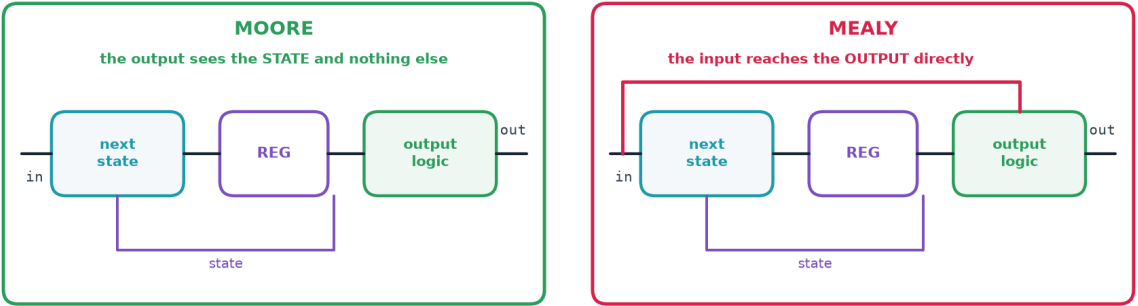
A naive latch rule flags every one of those if statements. Yosys builds a latch in none of them. make lintcheck puts the two opinions side by side on sixteen designs — and a linter that cries wolf gets switched off, which makes it worse than no linter at all.



Where the Output Comes From

Moore and Mealy: where the output comes from

One structural difference. Everything else follows from it.



	Moore	Mealy
output depends on	state only	state AND input
output appears	one cycle after the input	same cycle
output glitches	no - decoded from registers	yes - inherits the input's glitches
states needed	usually one more	usually one fewer
measured here	13 cells, 2 flops	14 cells, 2 flops
use it when	the output drives other logic or leaves the block	you cannot afford the cycle

Both machines accept exactly the same language

The lab proves it: one stream, one golden model, five matches, zero mismatches. They differ in WHEN they say so, not in WHAT they say.

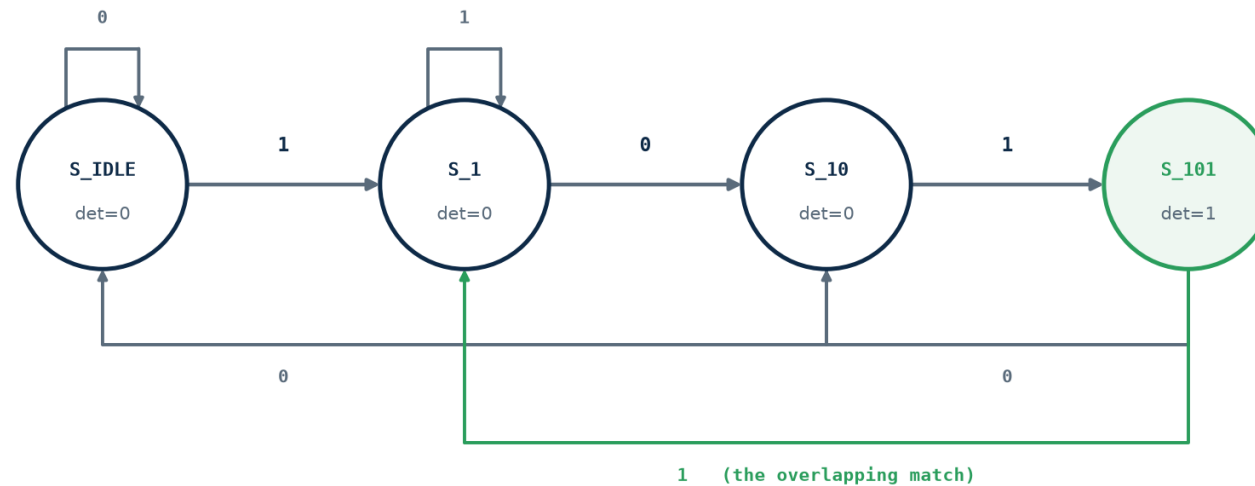
One structural difference — whether the output logic can see the input. Everything else follows.



The '101' Detector, Moore

The '101' detector as a MOORE state diagram

Four states. The output is written INSIDE the state, so it depends on nothing else.



The fourth state exists only to say 'a match just finished'

S_101 has the same successors as S_1 would after a 1, and as S_10 would after a 0 - it decides nothing new. It is there so that det can be decoded from the state register alone, which is the definition of Moore.

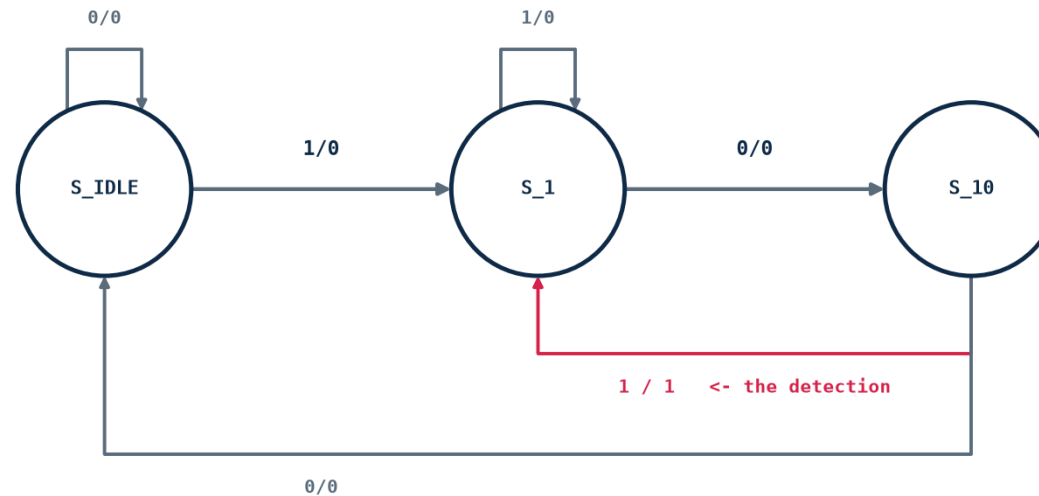
Four states. The output is decoded from the state register alone.



The Same Detector, Mealy

The same detector as a MEALY state diagram

Three states. Each arrow is labelled input / output.



The detection lives on an arrow, not in a state

'1/1' means: with the machine in S_10 and a 1 arriving, assert det NOW and go to S_1. One state fewer, one cycle sooner - and an output that moves the instant the input does, glitches included.

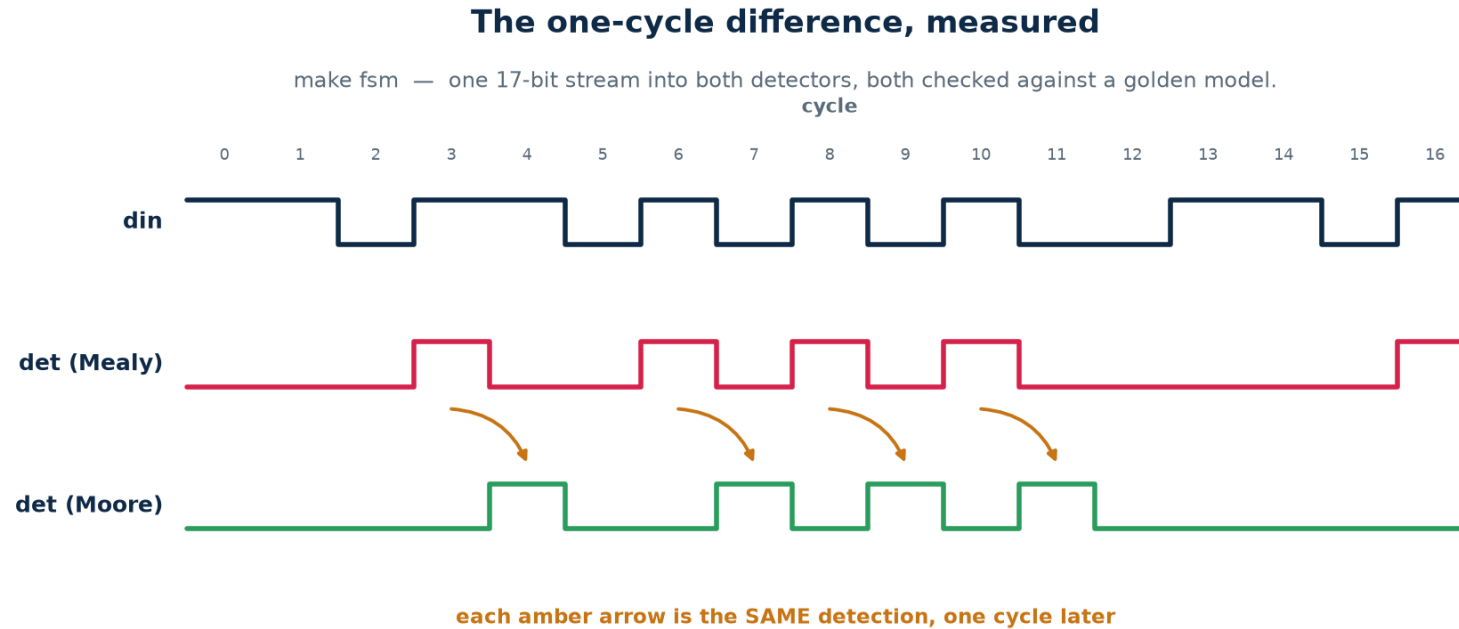
Both diagrams describe the same language. Neither is more correct than the other.

Three states. The output is written on the arrows, as input / output.



3.3 · MOORE AND MEALY

The One-Cycle Difference, Measured



5 matches in the stream · 0 mismatches against the golden model · Moore trails Mealy by exactly one cycle, every time

The golden model is computed from the STREAM, not from either machine - so a bug in one of them shows up as a mismatch rather than as two machines confidently agreeing on the wrong answer.

That is the whole point of writing a reference model by hand.

Five matches, zero mismatches against a golden model computed from the stream itself.



Choosing Between Them

CHOOSE MOORE

when the output leaves the block, drives other logic, or is timing critical. It comes out of a decode of registered bits, so it is clean, predictable and easy to constrain.

CHOOSE MEALY

when the cycle genuinely matters — a handshake that must be answered immediately, or a pipeline where one more cycle of latency costs throughput you cannot spare.

EITHER WAY, REGISTER IT AT THE BOUNDARY

A Mealy output that leaves your block hands the next designer a combinational path they did not ask for. If it crosses a module boundary, put a flip-flop on it — which turns it back into Moore.

The trap in the folklore

"Mealy needs fewer states" is true and almost never the reason to pick it. Here it saved one flip-flop's worth of nothing — both machines needed two — and cost one cell more. The real trade is the cycle against the glitch-free output, and that is a system question, not a coding one.



The Choice You Make By Typing Numbers

State encoding: the choice you make by typing numbers

The same four states, two ways of numbering them. Same behaviour, different hardware.

state	binary	one-hot
S_IDLE	2'b00	4'b0001
S_1	2'b01	4'b0010
S_10	2'b10	4'b0100
S_101	2'b11	4'b1000

encoding	flops	cells	what it buys
binary	2	13	shortest register
one-hot	4	30	shortest decode path

measured by make fsm on this toolchain

Read the measurement before repeating the folklore

On this generic gate library one-hot is BIGGER: four states need four flip-flops instead of two, and the next-state logic has to drive four bits instead of two. 30 cells against 13.

One-hot is still usually right on an FPGA. Both statements are true; neither is a rule.

encoding	the property	where it wins
binary	fewest flip-flops	ASIC, many states, area-critical
one-hot	one flop per state, decode is one wire	FPGA, speed-critical, few states
gray	one bit changes per legal transition	the state crosses a clock domain

In Verilog the encoding is yours; in VHDL it is the tool's unless you say otherwise

localparam S_IDLE = 2'd0 IS the encoding - you wrote the bits. type state_t is (S_IDLE, ...) is a set of NAMES, and the synthesiser picks the bits unless an attribute tells it otherwise.

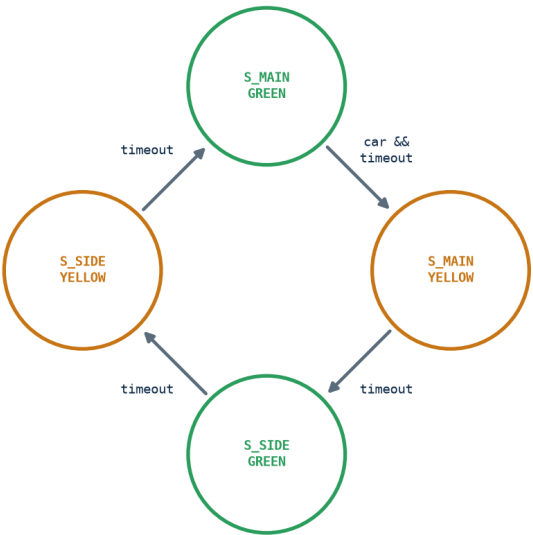
Same behaviour, different hardware — and the direction of the difference is not the one usually quoted.



The Traffic Light

A Moore controller with a datapath timer

fsm/traffic.v — a two-road junction with a car sensor on the minor road.



state	MAIN	SIDE	cycles
S_MAIN_GREEN	GREEN	RED	6
S_MAIN_YELLOW	AMBER	RED	2
S_SIDE_GREEN	RED	GREEN	6
S_SIDE_YELLOW	RED	AMBER	2

two properties, checked EVERY cycle

- P1 the two roads are never both green
- P2 a green never goes straight to red

40 cycles checked, 0 violations

Where the timer lives, and why it is not a state

A green phase lasts six cycles. Six states would be absurd, and six hundred would be impossible - so the count lives in a DOWN-COUNTER, and the state machine only ever asks it one question: are you at zero yet? That is a datapath and a controller again, in the smallest design in this topic.

Measured: 75 cells, 10 flip-flops - 2 for the state, 8 for the timer.

Two safety properties, checked on every one of forty cycles. Zero violations.



Checking a Property Instead of a Waveform

```
// P1 : the two roads are never green at the same time
if (main_light == GREEN && side_light == GREEN) begin
    $display("*** P1 VIOLATED at cycle %0d", cycles);
    errors = errors + 1;
end

// P2 : a green never goes straight to red - yellow comes first
if (prev_main == GREEN && main_light == RED) begin
    $display("*** P2 VIOLATED at cycle %0d", cycles);
    errors = errors + 1;
end
```

You write the rule once; the simulator tests it every cycle of every run

This is the cheap half of formal verification, available in any simulator, in about ten lines. It scales to runs far longer than anyone would page through by hand, and it keeps working after you change the design.

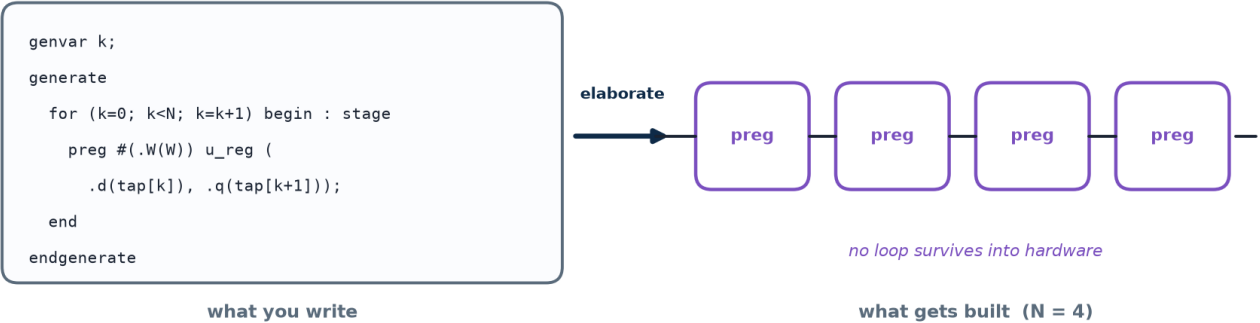
The expensive half — proving the property holds for every possible input — is what make prove does with a SAT solver.



Parameters, Hierarchy and Generate

From module to IP: parameters, hierarchy, generate

Reusable means ONE file that covers a family of widths and depths.



N	cells	flip-flops
1	8	8
2	16	16
4	32	32
8	64	64

Eight flip-flops per stage, exactly N stages, measured at four depths

parameter, hierarchy and generate are all instructions to the ELABORATOR. None of them exists in the netlist; they are gone before synthesis sees its first gate.

Eight flip-flops per stage, exactly N stages, measured at four depths.



What Elaboration Actually Does

```
module delayline #(parameter W = 8, parameter N = 4)
    (input clk, rst, en, input [W-1:0] din,
     output [W-1:0] dout);

    wire [W-1:0] tap [0:N];    // one more element than stages
    assign tap[0] = din;

    genvar k;
    generate
        for (k = 0; k < N; k = k + 1) begin : stage
            preg #(.W(W)) u_reg (.clk(clk), .rst(rst), .en(en),
                                .d(tap[k]), .q(tap[k+1]));
        end
    endgenerate

    assign dout = tap[N];
endmodule
```

The for loop is an instruction to the ELABORATOR, not a loop in hardware. After elaboration there are N preg instances and no loop anywhere.



Eight Questions

#	Question	The answer in one line
1	What are the two halves of almost every block?	a datapath that holds, a controller that decides
2	What crosses between them?	control signals down, status signals up
3	Name the three blocks of an FSM.	state register, next-state logic, output logic
4	Why does a default assignment prevent a latch?	every output is written before any branch runs
5	What does the output logic read in a Moore machine?	the state, and nothing else
6	How much later does a Moore output appear?	exactly one clock cycle, every time
7	One-hot: fewer or more flip-flops than binary?	more — one per state instead of $\log_2(\text{states})$
8	Does a generate loop exist in the netlist?	no — it is elaborated away before synthesis

Theory 4 is about the notation — which turns out to be the least important part.

Introduction to Hardware Description Languages

The syllabus says "HDLs such as Verilog or VHDL". The word doing the work is OR.

- Why an HDL is not a programming language
- Everything happens at once
- The parts of a module, and the word reg that confuses everyone
- Verilog and VHDL as reference cards, and a line-for-line mapping
- The anatomy of a testbench that checks itself
- Both languages actually run, and their transcripts diffed



An HDL Is Not a Programming Language

An HDL is not a programming language

It looks like one. That resemblance is the single biggest source of beginner bugs.

A PROGRAM

- statements run one after another
- a variable holds a value
- a loop repeats over time
- a function is called and returns
- you describe a PROCEDURE

AN HDL DESCRIPTION

- every block runs ALL THE TIME
- a signal is a wire with a value on it
- a loop is unrolled into hardware
- a module is INSTANTIATED and exists
- you describe a STRUCTURE

The sentence to keep in your head

You are not writing instructions for a machine to follow. You are writing a DESCRIPTION of a machine, which a tool will then build. The text is a blueprint, not a recipe.

Which is why "it compiles" means almost nothing, and why a construct can be perfectly legal Verilog and still have no hardware meaning at all.

You are not writing instructions for a machine to follow. You are writing a description of a machine.



Everything Happens At Once

Everything happens at once

The order you write the blocks in has no meaning whatsoever.

as written

```
assign p = a & b;
```

```
assign q = p | c;
```

```
assign r = q ^ d;
```

as built

AND

OR

XOR

*all three
exist at once,
for ever*

Write those three lines in any order you like

The circuit is identical. Put the r line first and it still works, because you did not write a sequence - you wrote three facts about three pieces of hardware that all exist together.

The one place order DOES matter

Inside a single always block using blocking assignments (`=`), statements run in order, like software. That is exactly why mixing `=` and `<=` in one block is confusing enough to be a lint rule.

Write those three assign lines in any order you like — the circuit is identical.



The Parts of a Verilog Module

The parts of a Verilog module

```
module counter #(parameter WIDTH = 4) (name, and parameters that let it be reused at other sizes
    input          clk, ports: the interface
    input          rst,
    output reg [WIDTH-1:0] count reg here means 'assigned in an always block', NOT a flip-flop
);

    always @(posedge clk) begin a clocked block: this is where flip-flops come from
        if (rst) count <= 0;
        else    count <= count + 1;
    end

endmodule
```

The word that confuses everyone: reg

reg does NOT mean register. It means "this signal is assigned inside a procedural block".
A reg assigned in an always @* block becomes pure combinational logic.

What actually creates a flip-flop is assigning inside always @(posedge clk) - nothing else.
SystemVerilog renamed reg to logic precisely to end this confusion.

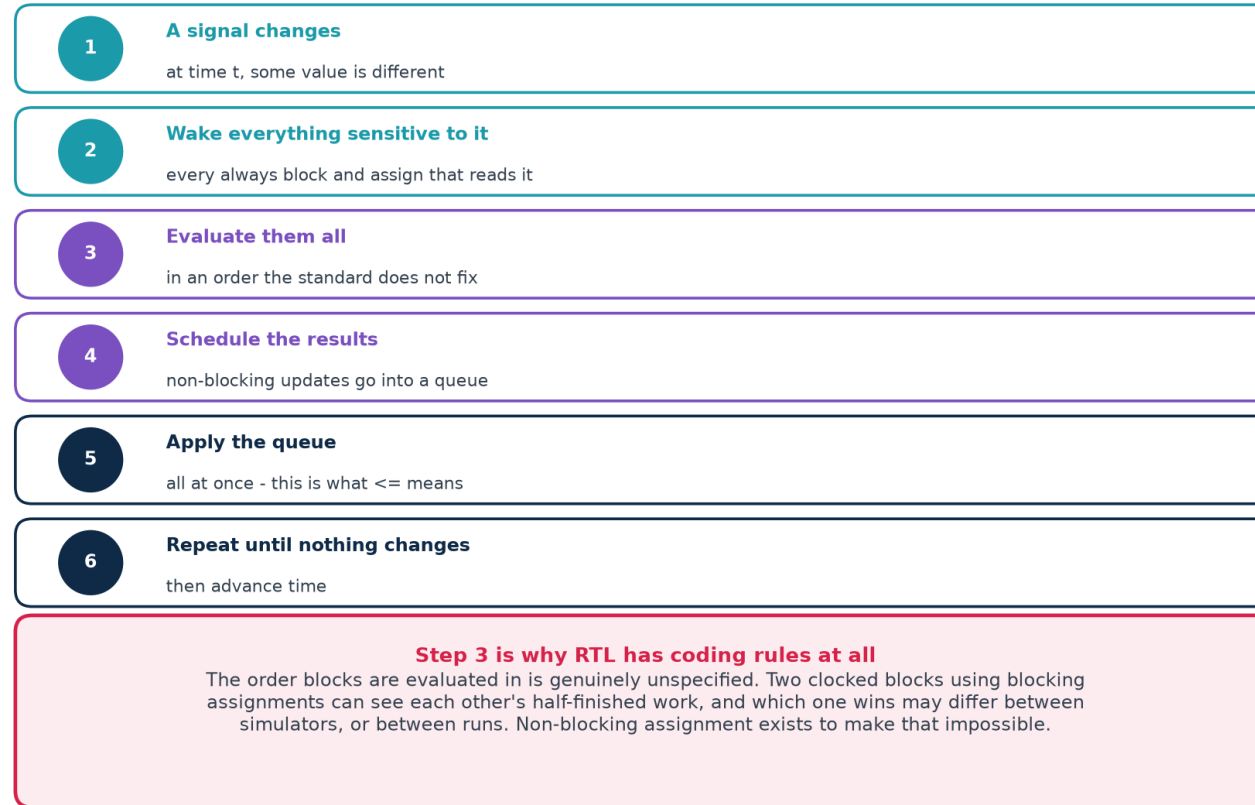
reg does not mean register. What creates a flip-flop is assigning inside always @(posedge clk) — nothing else.



How a Simulator Runs an HDL

How a simulator runs an HDL

Nothing runs continuously. The simulator jumps from event to event.



The order blocks are evaluated in is genuinely unspecified. That is why non-blocking assignment exists.



Verilog On One Page

Verilog: everything you need for this topic on one page

Not the whole language - the part that synthesises, which is much smaller.

THE MODULE

```
module counter4 (  
    input      clk,           a 1-bit input  
    input      rst_n,  
    output reg [3:0] count,    a 4-bit output you assign  
    output      tc             in a procedural block  
);  
    ...  
endmodule                    no semicolon after endmodule
```

NETS AND VARIABLES

```
wire y;                      driven by assign or a port  
reg q;                       assigned in always/initial  
reg [7:0] bus;               a vector, MSB first  
reg [7:0] mem [0:255];       an array of vectors  
localparam W = 8;           a constant, module-scope  
parameter N = 4;            a constant the caller can set  
  
`reg` does NOT mean register. It means 'assigned procedurally'
```

THE THREE BLOCKS

```
assign y = a & b;             continuous - combinational  
  
always @(*) begin             combinational  
    y = a & b;                 blocking =  
end  
  
always @(posedge clk) begin   sequential  
    q <= d;                   non-blocking <=  
end
```

OPERATORS WORTH KNOWING

```
& | ^ ~                      bitwise  
&& || !                       logical - result is 1 bit  
== !=                         equality (x/z make it x)  
=== !==                       identity - matches x and z too  
{a, b}                        concatenation  
{4{a}}                        replication - four copies of a  
a[3:1]                        part select  
cond ? x : y                  a multiplexer, written short
```

The synthesisable subset is roughly this page

make subset measures where the boundary actually is: eleven constructs, three of which are refused or silently ignored.

Not the whole language — the part that synthesises, which is much smaller.



VHDL On One Page

VHDL: the same ideas, spelled differently

Everything on this page has an exact counterpart on the Verilog page.

LIBRARIES AND ENTITY

```
library ieee;
use ieee.std_logic_1164.all;
use ieee.numeric_std.all;

entity counter is
    generic (WIDTH : positive := 4);
    port (
        clk : in std_logic;
        cnt : out std_logic_vector(3 downto 0)
    );
end entity counter;
```

Verilog has none to declare
std_logic, and 'or'
unsigned, signed, arithmetic
the INTERFACE only
= parameter

ARCHITECTURE

```
architecture rtl of counter is
    signal c : unsigned(3 downto 0);
begin
    process (clk)
    begin
        if rising_edge(clk) then
            c <= c + 1;
        end if;
    end process;
    cnt <= std_logic_vector(c);
end architecture rtl;
```

the IMPLEMENTATION
declared BEFORE begin
= always @(posedge clk)
<= is the signal assignment
explicit type conversion

TYPES - THE REAL DIFFERENCE

```
std_logic
std_logic_vector
unsigned / signed
integer
type state_t is (S_IDLE, S_1);

You must convert explicitly.
Verilog would just let you add.
```

one bit: 0 1 X Z U W L H -
a bundle of bits, no arithmetic
the SAME bits, with arithmetic
for constants and loop counters
a real enumerated type
and the compiler will make you

WHAT VHDL CATCHES THAT VERILOG DOES NOT

a non-exhaustive case
an illegal state value
a width mismatch
mixing signed and unsigned

will not analyse
not a member of the type
8 bits into 4 is an error
an error, not a surprise

The cost is keystrokes.
The benefit is that the bug
is found in the analyser,
not in the lab.

make langs - both languages, two simulators, transcripts diffed line by line

The counter agrees over 18 cycles; the '101' detector agrees over 17. Same designs, different notation.



Verilog to VHDL, Line For Line

Verilog to VHDL, line for line

Learn one and you can read the other in an afternoon. This table is the afternoon.

Verilog	VHDL	the difference that matters
module ... endmodule	entity + architecture	VHDL splits interface from implementation
input / output	port (... : in / out ...)	
parameter	generic	
localparam	constant	
wire	signal	VHDL has one kind, not two
reg	signal (or variable)	'reg' never meant register
[7:0]	(7 downto 0)	or (0 to 7) if you insist
always @(posedge clk)	process (clk) + rising_edge(clk)	
always @(*)	process (all inputs)	or process(all) in VHDL-2008
<= in a clocked block	<= signal assignment	same symbol, same idea
= in a comb block	:= variable assignment	different symbol
case ... default	case ... when others	'when others' is compulsory
localparam S = 2'd0	type state_t is (S, ...)	a number against a type
generate for	for ... generate	
\$display	report / write	
`timescale	units are written on the value	5 ns, not a directive

The syllabus says "HDLs such as Verilog or VHDL". The word doing the work is OR.

Nothing in this topic depends on which one you use. The abstraction levels, the synthesisable subset, the latch rules, the three-block FSM pattern and the whole flow are identical. Only the spelling changes.

Learn one and you can read the other in an afternoon. This table is the afternoon.



What a Testbench Is Made Of

What a testbench is made of

Topic 5 goes into verification properly. This is the minimum you need to finish Topic 2.

1 CLOCK	<pre>always #5 clk = ~clk;</pre> <i>a free-running clock - nothing else generates it</i>
2 RESET	<pre>rst_n = 0; @(negedge clk); rst_n = 1;</pre> <i>released away from the sampling edge, on purpose</i>
3 STIMULUS	<pre>din = stream[i];</pre> <i>drive on the inactive edge, sample before the active one</i>
4 GOLDEN MODEL	<pre>exp[i] = (s[i-2] & ~s[i-1] & s[i]);</pre> <i>computed from the STIMULUS, never from the DUT</i>
5 CHECKER	<pre>if (det != exp[i]) errors = errors + 1;</pre> <i>the line that makes it self-checking</i>
6 VERDICT	<pre>if (errors==0) \$display("PASS"); else ...</pre> <i>one line a human can read without a waveform</i>

A testbench without step 4 is not a test

If the expected answer comes from the design, the test passes whatever the design does. Every checker in this lab computes its expectation from the stimulus - which is why 'blocking version: 6 wrong cycles' could be printed at all.

Eyeballing a waveform is not step 5 either. It does not scale past about ten cycles, and it never runs twice.

Topic 5 covers verification properly. These six parts are the minimum you need to finish Topic 2.



4.6 · TWO LANGUAGES

Verilog and VHDL, Side By Side

Verilog and VHDL: the same ideas, different notation

	Verilog	VHDL
origin	1984, Gateway Design Automation	1983, US Department of Defense
standard	IEEE 1364, then 1800 (SystemVerilog)	IEEE 1076
feel	terse, C-like	verbose, Ada-like
typing	weak - it will let you do most things	strong - explicit conversions required
case sensitive	yes	no
interface / body	one module	entity and architecture, separately
a clocked block	always @(posedge clk)	process (clk) ... if rising_edge(clk)
assignment	<= and =	<= and :=
used most in	North America, ASIC flows	Europe, defence, aerospace

None of those differences are about hardware

Both describe registers, combinational logic and hierarchy. Both synthesise to the same gates.
An engineer who understands RTL can read the other one after an afternoon; an engineer who has only memorised syntax can read neither.

The syllabus says "Verilog or VHDL". The word doing the work is OR.

None of those differences are about hardware. Both describe registers, logic and hierarchy; both synthesise to the same gates.



Both Designs, In Both Languages, Actually Run

The same designs, in both languages, actually run

Verilog · iverilog

```
always @(posedge clk)
  if (rst) count <= 0;
  else if (en)
    count <= count + 1;
  assign tc = &count;
```

VHDL · ghdl

```
if rising_edge(clk) then
  if rst = '1' then
    cnt <= (others => '0');
  elsif en = '1' then
    cnt <= cnt + 1;
```

design	Verilog	VHDL	cycles	diff
counter	rtl/counter.v	vhdl/counter.vhd	18	IDENTICAL
'101' detector	fsm/seq101_moore.v	vhdl/seq101_moore.vhd	17	IDENTICAL

Two languages, two simulators, transcripts diffed

Not "they look similar". The logs were compared line by line by diff, and there was nothing to report.

The state machine is the interesting one: VHDL gives the states a real enumerated TYPE, so an illegal state will not compile. Verilog gives them numbers.

Not "they look similar"

Both transcripts were compared line by line by diff, and there was nothing to report. The state machine is the interesting one: VHDL gives the states a real enumerated TYPE, so an illegal state will not compile. Verilog gives them numbers.



Which HDL Should You Learn?

Which HDL should you learn?

Learn the **CONCEPTS**. The notation follows in an afternoon.

registers and combinational logic · the clock edge · reset · hierarchy
blocking against non-blocking · the synthesisable subset · what a tool will build

if you are doing	use	because
This course	Verilog	it is what Module 2 and the labs use
Most ASIC work	SystemVerilog	Verilog plus better types and verification
Much European and defence work	VHDL	strong typing, long project lifetimes
Increasingly	both	large projects mix them in one flow

The lab proves the point rather than asserting it

The same counter, written in both, simulated by two different simulators, produced identical transcripts over 18 cycles.

Learn the concepts. The notation follows in an afternoon.



THEORY 4 · CHECKPOINT

Seven Questions

#	Question	The answer in one line
1	Name the biggest difference between an HDL and a program.	everything in an HDL runs all the time, concurrently
2	Does the order of assign statements matter?	no — they describe hardware that exists together
3	What does reg actually mean?	assigned in a procedural block; NOT a flip-flop
4	Why is the evaluation order unspecified?	the standard leaves it free; <= makes it not matter
5	Verilog or VHDL?	either — the concepts are identical, and the lab proves it
6	What does VHDL catch that Verilog does not?	illegal states, non-exhaustive cases, width mismatches
7	Where must a testbench's expected answer come from?	the stimulus — never from the design under test

PRACTICAL

Labs A-N

26 hours, 103 graded exercises, and not one number in this deck that you cannot reproduce yourself.

- Which tool answers which question, and how to install them
- Labs A-D: what RTL means, the ladder, proof, and the subset
- Labs E-H: the mismatch, the coding rules, style, and the two pitfalls
- Labs I-L: state machines, a controller with a timer, datapath and controller, and reuse
- Labs M-N: two languages, the whole flow, and the vendor tools



The Tools, and What Each One Is For

The tools, and what each one is for

tool	licence	what it does	used in
Icarus Verilog	free	simulate Verilog	this lab
GHDL	free	simulate VHDL	this lab
GTKWave	free	look at the waveforms	this lab
Yosys	free	synthesise, and prove equivalence	this lab
Verilator	free	fast simulation, and a good linter	recommended
rtl_lint.py	you write it	the seven coding rules	this lab
Vivado	free WebPACK	synthesis and implementation, Xilinx	syllabus tool
ModelSim / Questa	free starter	simulation, both languages	syllabus tool
Design Compiler	commercial	ASIC synthesis	industry

Everything in this topic runs on the free rows

Yosys is the interesting one: it is the only free tool here that will both synthesise your RTL and PROVE that the netlist it produced is equivalent to what you wrote.

Vivado and ModelSim are named by the syllabus and are covered in the deck; they were not used to produce any number in these materials.

Yosys is the interesting one: the only free tool here that both synthesises your RTL and PROVES the netlist matches it.



Two apt Lines, and the Whole Lab Runs

Install this, and the whole lab runs

Two apt lines. No licence, no account, no download manager.

REQUIRED

Debian / Ubuntu / WSL2

```
sudo apt update

sudo apt install yosys iverilog gtkwave python3

sudo apt install ghdl      # only for the VHDL comparison

Icarus  simulates Verilog - and the transistor level too

GHDL    simulates VHDL, so 'Verilog or VHDL' can be shown

Yosys   synthesises, and proves equivalence with a SAT solver

GTKWave for when the transcript is not enough
```

```
yosys -V && iverilog -V && ghdl --version
```

*Three version strings and every target in the lab Makefile will run.
Without ghdl, everything except make langs still works.*

Debian / Ubuntu / WSL2

```
sudo apt install yosys iverilog gtkwave python3
sudo apt install ghdl      # only for the VHDL comparison
yosys -V && iverilog -V && ghdl --version
```

Without ghdl everything except make langs still runs.



The Vendor Tools the Syllabus Names

The vendor tools the syllabus names

	what to do	watch out for
1	Create a free AMD/Xilinx account	the download will not start without one
2	Download the Unified Installer (~200 MB web installer)	not the 40 GB full archive
3	Choose Vivado ML Edition, then ML Standard	Standard is free; no licence file needed
4	Deselect every device family but the one you use	cuts 40 GB to about 8 GB
5	ModelSim: Intel FPGA Starter Edition is free	simulates both Verilog and VHDL
6	Verify: vivado -version and vsim -version	both then run headless

What these add, and what they do not

They add a real device library, real timing numbers, and a flow you will meet at work. They do not add anything to the CONCEPTS in this topic - RTL is RTL, and the free toolchain shows all of it.

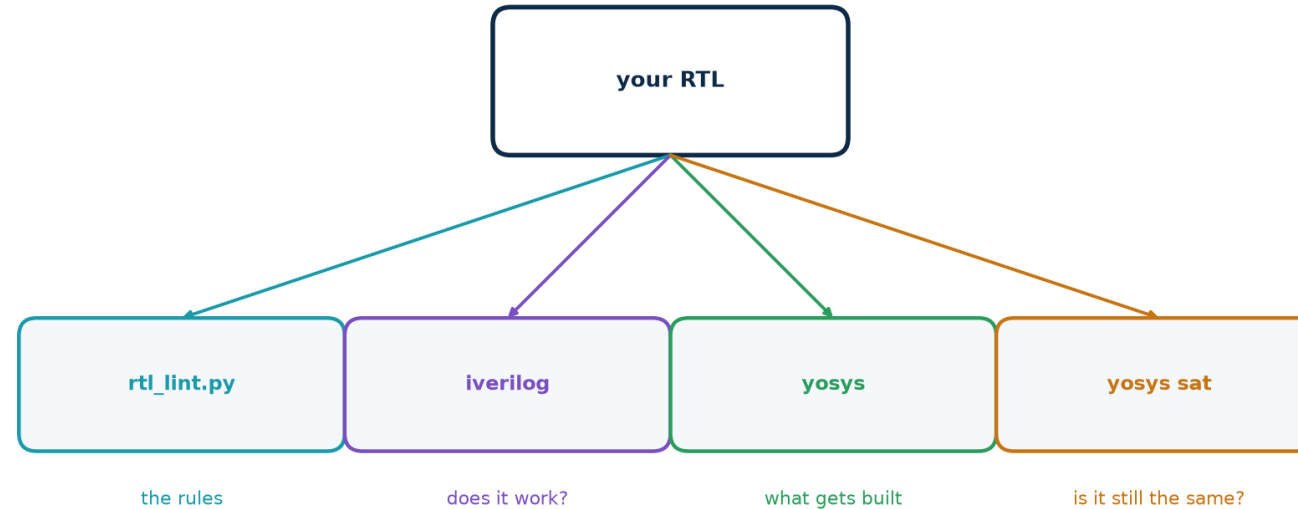
Learn the ideas on the free tools, where the whole flow takes seconds. Then run the same design through Vivado and recognise every stage.

They add a real device library and a flow you will meet at work. They add nothing to the concepts.



Four Questions About One Piece of Code

What the lab does with your RTL



make lint make ladder make subset make prove make flow

Four different questions about one piece of code

Does it follow the rules? Does it do what the spec says? What will actually be built from it?
And is that thing still the design you wrote?

A methodology is exactly the discipline of asking all four, every time.

Does it follow the rules? Does it do what the spec says? What will be built? Is that still your design?



Fourteen Parts, Twenty-Six Hours

The practical component - 26 hours, fourteen parts

Module 2 practical: RTL Design and Implementation Labs (40 h). This is Topic 2's share.

A	What RTL means	registers, transfers, and the clock edge	1 h
B	The abstraction ladder	one adder, four levels, all simulated	2 h
C	Proof, not just testing	equivalence checking, and a bug it catches	2 h
D	The synthesisable subset	eleven constructs, measured	2 h
E	Simulation vs silicon	the sensitivity-list mismatch	1 h
F	The coding rules	build the linter, then check the linter	2 h
G	Coding style	one function three ways: 3, 6 and 10 cells	1 h
H	The two pitfalls	the inferred latch, and blocking in a clocked block	2 h
I	State machines	Moore, Mealy, encoding - all measured	3 h
J	A controller with a timer	the traffic light, and property checking	2 h
K	Datapath and controller	the accumulator, and its control bundle	2 h
L	From module to IP	parameters, hierarchy, generate	2 h
M	Two languages	counter and FSM, Verilog and VHDL, diffed	2 h
N	The flow and the vendor tools	spec to formal proof, then Vivado	2 h

103 graded exercises, every one with a worked solution

A-D build the mental model. E-H are the methodology, made mechanical. I-L are the patterns every real block is built from. M-N connect it to other languages and to the tools you will meet at work.

Every number in this deck came from a target in that lab.



What RTL Means, the Ladder, and Proof

A • what RTL means (1 h)

Run make transfer. Follow a single 5 through four registers.

Predict the value in each register on each cycle BEFORE running it.

Deliverable: the table, and why y is 1 on cycle 0.

B • the ladder (2 h)

Write the same full adder at all four levels.

Run make ladder — all four together, exhaustively.

Then synthesise each and compare cell counts.

Deliverable: why behavioural won.

C • proof (2 h)

Run make prove. Three pairs proved equivalent.

Then break the adder yourself and watch the checker catch it.

Deliverable: why 8 patterns was enough here and will not be next time.

The exercise that teaches the most is C

Delete a different term from fa_broken.v and predict which input pattern the solver will find. Getting that prediction right means you understand both the circuit and the tool.



The Subset, the Mismatch, and the Rules

```
$ make subset
s03_latch      OK      1      YES      inferred a LATCH: $_DLATCH_P_
s08_whileloop REFUSED -      -      "only allowed in constant functions"
s10_divide     OK      371    no      a full combinational divider
s11_shift     OK      0      no      no logic at all - just wires

$ make mismatch
change b (list ASLEEP)  a=1 b=1  RTL y=0  NETLIST y=1  <-- THEY DISAGREE

$ make lintcheck
10 files · linter and Yosys agree on every one · 0 disagreements
```

Exercise	Task
D.1	Predict the cell count for each construct before running make subset
D.2	Add a twelfth construct of your own and predict the result
E.1	Explain the mismatch to someone who has not seen it, in three sentences
F.1	Add rule L008 to the linter — your choice — and justify it
F.2	Find a file the linter gets wrong, and say why regexes cannot fix it



Coding Style, and the Two Pitfalls

```
$ make mux
mux4_assign vs mux4_case      EQUIVALENT (proved, 92 SAT variables)
mux4_assign                   3 cells    0 flip-flops
mux4_if                       6 cells    0 flip-flops
mux4_case                    10 cells    0 flip-flops

$ make pitfalls
non-blocking version : 0 wrong cycles
blocking version    : 6 wrong cycles
shift_nb (non-blocking)  3 cells    3 flip-flops
shift_bl (blocking)     1 cells    1 flip-flops
s03_latch (missing else) 1 cells    *** 1 LATCH INFERRED ***
```

Exercise	Task
G.1	Predict the three cell counts before you run make mux
G.2	Write a fourth style of your own and predict where it lands
H.1	Predict q_bl for the first six cycles, then run it
H.2	Explain in two sentences why no tool warned about shift_bl
H.3	Add an else to s03_latch and confirm the latch disappears



State Machines, a Timer, and a Datapath

```
$ make fsm
matches in the stream : 5      mismatches vs golden : 0
PASS - same language, Moore trails Mealy by one cycle,
      and the one-hot encoding is indistinguishable
seq101_moore  binary      13 cells    2 flip-flops
seq101_moore  one-hot     30 cells    4 flip-flops
cycles checked : 40    property violations : 0      (traffic light)

$ make dpctrl
golden total : 157    hardware sum : 157
accum_ctrl   (controller)    10 cells    2 flip-flops
accum_datapath (datapath)    145 cells   24 flip-flops
```

Exercise	Task
I.1	Draw the state diagram for '110' before you write any code
I.2	Write it in both styles and confirm the one-cycle offset
I.3	Re-encode the Moore machine as gray and measure the cells
J.1	Add a pedestrian request input and a third property to check
K.1	Move the terminal-count test into the controller. What breaks, and why is it worse?



From a Module to an IP

```
$ make reuse
delayline mismatches : 0          prescaler ratio errors : 0

delayline #(W=8, N) synthesised at four different depths:
N = 1          8 cells      8 flip-flops
N = 2          16 cells     16 flip-flops
N = 4          32 cells     32 flip-flops
N = 8          64 cells     64 flip-flops
```

Exercise	Task
L.1	Predict the cell count for $N = 16$ before running it
L.2	Add a bypass parameter that removes the registers entirely when $N = 0$
L.3	Instantiate counter_n three times to divide by 4096, and check the ratio

L.2 is the one worth arguing about

A generate loop with $N = 0$ builds nothing, so dout must come straight from din. Getting that right means understanding that the parameter is resolved before any hardware exists — which is the whole point of the lab.



Two Languages, the Whole Flow, and the Vendor Tools

```
$ make langs
=== Verilog: iverilog ===      === VHDL: ghdl ===
IDENTICAL over all 18 cycles - including the wrap and the terminal count.
IDENTICAL over all 17 cycles, detections included.  (the '101' detector)

$ make flow
STAGE 2  LINT                0 issues
STAGE 4  SYNTHESIS           12 cells
STAGE 6  COMPARE             IDENTICAL on all 18 cycles
STAGE 7  PROVE               Equivalence PROVEN by induction

All seven stages passed.
```

Lab N: the same design in Vivado

Read the design, synthesise for a real part, report utilisation, write the netlist for gate-level simulation. Vivado is not installed in the environment these materials were built in, so lab N is the one whose output is not reproduced here — run it and record what you get.



The Same Flow in Vivado and ModelSim

The same flow in Vivado and ModelSim

1

`vlog counter.v ; vsim -c tb_counter`

ModelSim: compile and simulate - the same job as iverilog + vvp

2

`read_verilog rtl/counter.v`

Vivado: read the design

3

`synth_design -top counter -part xc7a35t`

synthesise for a real device

4

`report_utilization`

how many LUTs and flip-flops - the same question as 'cells'

5

`write_verilog -mode funcsim net.v`

the netlist, for gate-level simulation

Stated plainly: these commands were not run here

Vivado and ModelSim are not installed in the environment these materials were built in. Every number quoted in this deck came from the free toolchain, and is reproducible with make.

The commands above are from the vendor documentation. Run them and record what you get.

Stated plainly on the slide: these commands were not run here. Every number in this deck came from the free toolchain.



How the 103 Exercises Are Weighted

How the 103 exercises are weighted

part	exercises	weight	assessed on
A · what RTL means	6	5%	can you read a transfer as hardware
B · the ladder	9	8%	same circuit, four notations
C · proof	7	7%	why exhaustive testing stops working
D · the subset	10	9%	prediction before measurement
E · sim vs silicon	5	5%	explaining the mismatch precisely
F · coding rules	10	9%	the rule, and the reason for it
G · coding style	5	5%	equivalent is not identical
H · the two pitfalls	7	7%	what the code BUILDS, not what it says
I · state machines	12	13%	the three-block form, and Moore vs Mealy
J · controller + timer	6	6%	writing a property, not reading a waveform
K · datapath/controller	7	8%	where each signal belongs, and why
L · module to IP	6	6%	elaboration against synthesis
M · two languages	5	5%	reading VHDL without panic
N · the flow	8	7%	evidence at every stage

The rule that runs through every part

Predict, then measure. An exercise where you ran the command and wrote down the answer earns very little; one where you wrote the prediction first and then explained the difference earns it all.

Being wrong and knowing why is the whole point of a lab.

Predict, then measure. Being wrong and knowing why is the whole point of a lab.



What This Lab Does Not Show You

- `rtl_lint.py` is regular expressions, not a Verilog parser. It reads code the way a careful reviewer skims it and can be fooled by unusual formatting.
- The cell counts come from Yosys' generic library, not a foundry library. They compare designs against each other; they are not area figures.
- `fa_switch.v` is a functional transistor model. There are no threshold voltages and no capacitances — SPICE is the tool for that.
- Yosys accepts initial blocks because it targets FPGAs. An ASIC flow would not, and that difference is a genuine trap.
- Vivado and ModelSim were not run. Every measured number here came from iverilog, ghdl and yosys, and is reproducible with make.
- Nothing here covers verification methodology beyond a directed testbench and a handful of assertions — that is Topic 5, a large subject on its own.
- The state machines here are small and single-clock. Clock-domain crossing, reset synchronisers and safe-state recovery are all real, and all later.



SUMMARY

Topic 2 In Fourteen Lines

- RTL says which registers exist and what transfers into them. Nothing else.
- There are two kinds of logic, combinational and sequential, and no third.
- `<=` reads every right-hand side first, then updates together. `=` does not — and the blocking version built 1 flip-flop where 3 were meant.
- Four levels of abstraction describe the same circuit — proved, not assumed.
- Write at the highest level that says what you mean: behavioural gave 5 cells against 6.
- Not all legal Verilog is synthesisable, and `a / b` cost 371 cells against 0.
- A combinational block that does not assign on every path builds a latch; a default assignment at the top of the block prevents it.
- An incomplete sensitivity list makes simulation and silicon disagree silently.
- Equivalent is not identical: three mux styles, 3, 6 and 10 cells.
- Almost every block is a datapath and a controller. Here: 145 cells against 10.
- Write state machines in three blocks — register, next state, output.
- Moore trails Mealy by exactly one cycle, and one-hot was BIGGER here.
- `parameter`, `hierarchy` and `generate` are elaborated away before synthesis.
- Verilog and VHDL produced identical transcripts, on both designs.



CLOSE

What To Do Next

By the end of this subtopic you can

- ✓ Read an always block and say which registers it creates, and which logic sits between them.
- ✓ Write the same function at four levels of abstraction, and say what each level costs you.
- ✓ Choose the right level - and defend the choice with a cell count.
- ✓ Prove two descriptions are the same circuit, instead of hoping.
- ✓ Say which Verilog constructs synthesise, which do not, and which are traps.
- ✓ Recognise an inferred latch before the tool builds one.
- ✓ Explain why an incomplete sensitivity list is worse than an error.
- ✓ Apply the seven coding rules, and say why each one exists.
- ✓ Split a block into a datapath and a controller, and say which signals cross between them.
- ✓ Write a state machine in the three-block form, in either style, and justify the choice.
- ✓ Parameterise a design so one file covers a family of widths and depths.
- ✓ Read a VHDL design without needing it translated.
- ✓ Take a design from spec to formal proof and show your evidence at each stage.

And one habit to carry into every design you ever write

Before you write a line of RTL, know what you expect the tool to build. Then check whether it did.
Every bug in this topic lives in the gap between those two things.

Coming next in Module 2

Topic 3 covers the digital logic this all rests on. Topic 4 writes much more RTL. Topic 5 is verification — how you know it works. Topic 6 is timing — whether it is fast enough.